Shreyansh Modi

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Education

Indian Institute of Technology, Roorkee | *B.Tech in Electrical Engineering* *2024 – 2028 (Expected)*
CGPA: 7.573/10 (3.03/4)

Vishwa International Academy | Class XII (CBSE) — 93.20% *2024*

St. Teresa School | Class X — 97.20% *2022*

Experience

Machine Learning Engineer Intern | *Infigro.ai* *Aug 2025 – Oct 2025*

- Built a Data Analysis Agent capable of interpreting bar charts and trendline data for automated insight generation.
- Developed an LSTM-based time-series forecasting system for analysing betting trends and user behaviour.
- Integrated LLM API calls to convert chart data and model predictions into natural-language analytical insights.

Research Intern | *Electrical Engineering Department, IIT Roorkee* *Nov 2024 – Dec 2024*

- Implemented and tuned PID controllers for a DC motor setup to evaluate the impact of cyber-physical attacks on closed-loop control.
- Simulated signal injection and spoofing attacks on cyber-physical systems using MATLAB and Arduino.

Reviewer | *NeurIPS 2026* *2026*

Publications

Rethinking the Effectiveness of Contrastive Decoding in Mitigating Hallucinations in MLLMs
A. Bendre, S. Modi, G. Gupta, K. Grover, C. Aggarwal *VLM4RWD, NeurIPS'26*
Showed on LLaVA-1.5 with POPE and CHAIR that contrastive decoding is 94% greedy-equivalent with over 90% logit overlap, tracing its gains to a shift in decoding behaviour rather than hallucination mitigation.

Guidance for Low-Level Perceptual Editing in Unconditional Diffusion Models
S. Modi, A. Tomar, A. Aggarwal arXiv | Code *CVPR'26 GCV Workshop*
Extracted concept vectors from DDPM U-Net bottleneck activations to enable training-free perceptual image editing, steering reverse diffusion through latent-space interventions and classifier-free guidance for 13.9%, 7.76% and 6.07% FID improvements on contrast, saturation and sharpness.

How Many Counterfactuals Does It Take? Probing VLM Hallucinations Through Circuits and Causal Effects
A. Gupta, S. Singh, A. Sinha, S. Modi, A. Tomar arXiv | Code *arXiv Preprint, 2026*
Built an activation patching pipeline with causal influence scoring to localise hallucination-sensitive attention heads and MLP layers, evaluated on LLaVA-1.5 and Qwen3-VL with 91.8% POPE accuracy and gains on HallusionBench.

Projects

RE-UltraBreak: Universal Jailbreak Attacks on VLMs *Jun 2026 – Aug 2026*

- Conducted a reproducibility study on UltraBreak, auditing safety vulnerabilities in black-box VLMs by replicating the optimisation pipeline.
- Generated universal, cross-model adversarial jailbreaks using randomised image transformations and loss relaxations to maximise transferability.

VLM Hallucination Mitigation via Self-Verification Decoding *Jun 2025 – Sep 2025*

- Mitigated visual hallucinations in VLMs with a decoding-time framework that enforces strict textual grounding.
- Improved reliability and semantic alignment of multimodal outputs, validated on the POPE and CHAIR benchmarks.

Reproducibility Study: Diffusion Models Beat GANs | GitHub *Aug 2025 – Sep 2025*

- Replicated OpenAI's unconditional U-Net and noise-robust classifier framework to implement classifier guidance during reverse diffusion.
- Validated that scaling classifier gradients trades sample diversity for visual fidelity, shifting the generative distribution toward target classes.

Adobe Photo Editor | *Inter IIT Tech Meet 14.0, 4th place* | GitHub *Nov 2025 – Dec 2025*

- Built a computer vision photo application with automated, context-aware pipelines for background removal and style transfer.
- Optimised MobileSAM inference through ONNX Runtime and INT8 quantisation, enabling real-time edge deployment.
- Integrated typographic editing and texture transfer for style manipulation and graphic overlays.

InVisionDX: CNN-Based Disease Detection | GitHub *Mar 2025 – Apr 2025*

- Built a web-based medical image classification system using ResNet and VGG16 with over 95% classification accuracy.
- Trained custom CNN architectures to detect COVID-19, pneumonia, tuberculosis, lung cancer and Alzheimer's disease.
- Deployed the models with a Flask backend and an HTML/CSS/JavaScript frontend.

Technical Skills

Languages: Python, C++, SQL
Frameworks & Tools: PyTorch, NumPy, Pandas, Scikit-learn, OpenCV, Pillow, Torchvision, Hugging Face, Flask, Git
Domains: Computer Vision, Generative AI, Multimodal Learning, Large Language Models, Adversarial ML, VLM Security
Coursework: Machine Learning (CS229, Stanford), Deep Learning for Computer Vision (Michigan), Deep Learning Specialisation (DeepLearning.AI), Linear Algebra, Probability & Statistics, Optimisation

Positions of Responsibility

Core Member | *Data Science Group, IIT Roorkee* *Feb 2025 – Present*

- Mentored students for Build Your Own Project (BYOP) 2026 on project design and implementation, and organised a hands-on technical workshop to drive data science research on campus.